NAME: ATCHAYA VHARSNE S

REG NO: 192524185

COURSE CODE : CSA1515

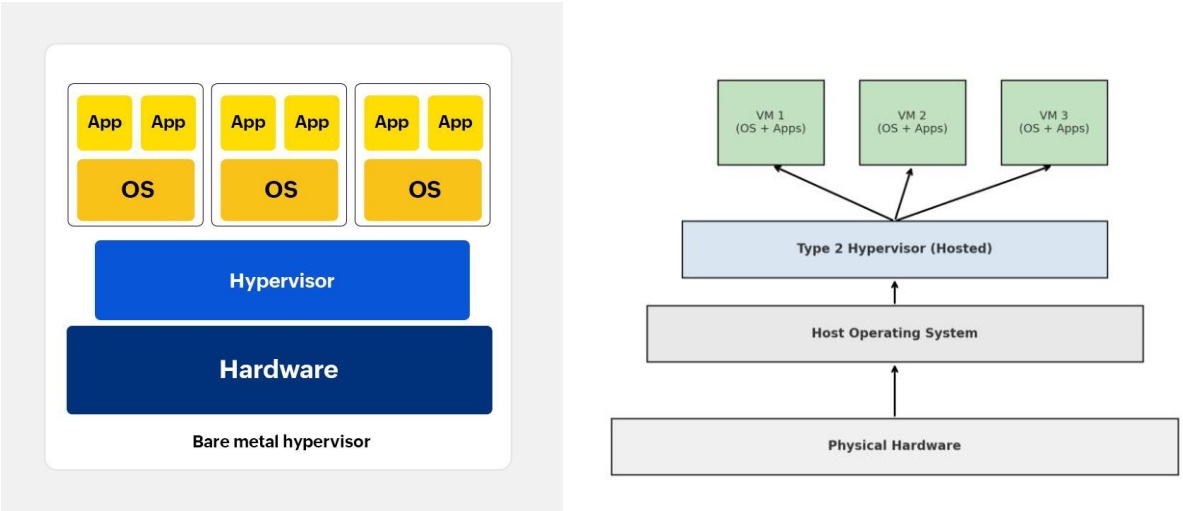
COURSE NAME: CLOUD COMPUTING AND BIG DATA ANALYTICS

CO2 AT1 Analytical Problem Solving

1. **Compare Type-1 and Type-2 hypervisors for a cloud data center hosting mission-critical applications. Analyse performance, security, and manageability, then justify the better choice.**

   A hypervisor is software or firmware that creates and manages virtual machines (VMs). It allows multiple operating systems to run on a single physical computer.

   There are two main types of hypervisors: Type-1 and Type-2.



   Type-1 Hypervisor
   Runs directly on physical hardware.

   Type-2 Hypervisor
   Runs on top of a host operating system.

Comparison table

Feature	Type-1	Type-2
Architecture	Runs directly on hardware	Runs on a host OS
Performance	Generally higher	Some additional overhead
Security	Smaller host-OS attack surface	Host OS adds another attack surface
Management	Requires specialized administration	Easier for beginners
Scalability	Suitable for large datacenters	Suitable for personal use and testing
Examples	VMware ESXi, Microsoft Hyper-V, Xen	Oracle VirtualBox, VMware Workstation

Analysis

Performance: Type-1 hypervisors generally provide better performance because they operate directly on hardware. They are suitable for applications that require high availability and predictable resource usage.

Security: Type-1 hypervisors reduce dependence on a general-purpose host OS. However, they still require security patches, access controls and proper configuration. Type-2 hypervisors depend on the security of both the host OS and the hypervisor.

Manageability: Type-2 hypervisors are easier to install and use on ordinary computers. Type-1 hypervisors offer centralized management features that are useful in enterprise datacenters.

Justification

For a cloud datacenter hosting mission-critical applications, a Type-1 hypervisor is the better choice because it generally offers better performance, efficient resource utilization and enterprise-level management.
Examples include VMware ESXi and Xen. Security should be strengthened through regular updates, role-based access control, network segmentation and continuous monitoring.

Conclusion

Both hypervisors support virtualization, but Type-1 hypervisors are more suitable for enterprise cloud environments, while Type-2 hypervisors are ideal for learning, development and testing.

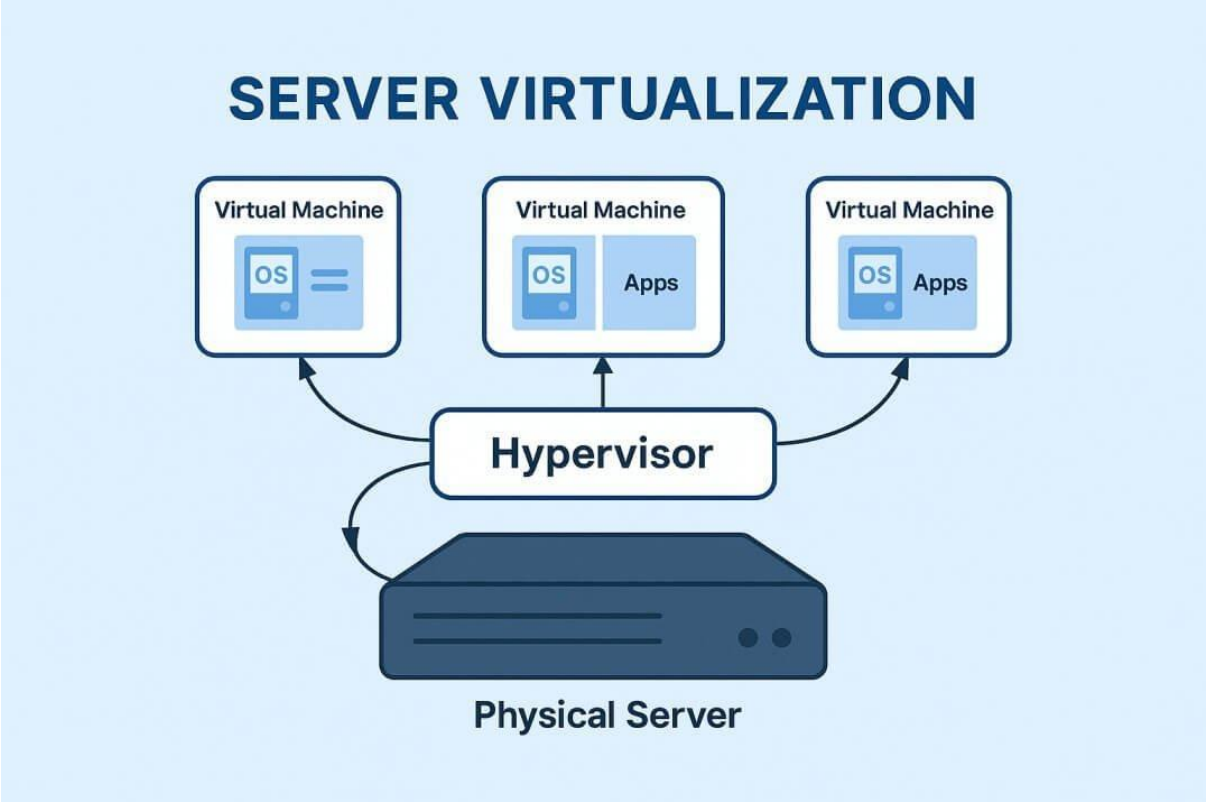
Question 2: Virtualization Isolation and Attack Surfaces (10 Marks)

Question: A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

1. Introduction

Virtualization is a technology that allows multiple virtual machines (VMs) to run on a single physical server. Each VM can run its own operating system and applications. In a cloud environment, finance, healthcare, and student portals can share the same physical hardware while keeping their data and applications separate through virtualization.

2. How virtualization supports isolation



The virtualization architecture consists of the following components:

1. Physical hardware: Provides the CPU, RAM, storage and network resources.
2. Hypervisor: Creates and manages VMs and allocates hardware resources to each VM.
3. Virtual machines: Each VM operates independently with its own operating system and applications.

4. Virtual networks: Separate network segments help prevent unauthorized communication between workloads.
5. Access control: Authentication and permissions ensure that only authorized users can access sensitive resources.

Example

Workload	Purpose	Security requirement
Finance portal	Handles payments and financial records	Strong access control and encryption
Healthcare portal	Stores patient information	Strict privacy and data protection
Student portal	Manages student details and academic records	Authentication and restricted access

Even though all three workloads use the same physical server, virtualization helps prevent one VM from directly accessing another VM's memory, files or resources.

3. Two major attack surfaces

A. Hypervisor attack surface

The hypervisor manages the virtual machines and controls access to physical resources. If an attacker exploits a vulnerability in the hypervisor, they may gain unauthorized access to multiple VMs.

Possible impact:

- Unauthorized access to sensitive data.
- Compromise of multiple virtual machines.
- Disruption of cloud services.

Preventive measures:

- Regularly update and patch the hypervisor.
- Restrict administrative access using multi-factor authentication.
- Monitor hypervisor logs and suspicious activities.
- Disable unnecessary services and interfaces.

B. Virtual machine and virtual network attack surface

Vulnerabilities in a VM's operating system, applications or virtual network configuration can allow attackers to compromise a workload or attempt to access other workloads.

Possible impact:

- Theft of financial or healthcare data.
- Malware spreading between systems.
- Unauthorized access to student records.
- Network attacks against other VMs.

Preventive measures:

- Install security updates on guest operating systems.
- Use firewalls and network segmentation.
- Apply least-privilege access controls.
- Encrypt sensitive data and monitor network traffic.

4. Recommended security architecture

The cloud provider should implement the following measures:

1. Use a secure, regularly patched Type-1 hypervisor.
2. Place finance, healthcare and student workloads in separate VMs.
3. Configure separate virtual networks and firewall rules.
4. Encrypt sensitive data both at rest and in transit.
5. Apply role-based access control and multi-factor authentication.
6. Continuously monitor logs and perform security audits.

5. Conclusion

Virtualization allows finance, healthcare and student workloads to share physical hardware while maintaining logical isolation. However, vulnerabilities in the hypervisor and VM or virtual network components can threaten this isolation. Therefore, secure configuration, regular patching, network segmentation, encryption and continuous monitoring are essential for protecting cloud environments.

Question 3: VM Host Utilization and Bottleneck Analysis (10 Marks)

Question: Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

1. Introduction

Virtual machines (VMs) share the physical resources of a host server, such as CPU, memory and storage. When these resources become overloaded, VM performance decreases and applications may respond slowly.

The given utilization snapshot helps identify possible performance bottlenecks and suitable corrective actions.

2. VM host utilization snapshot

Parameter	Observed value	Analysis
CPU utilization	92%	Very high; possible CPU bottleneck
Memory utilization	88%	High; limited available memory
Storage I/O latency	35 ms	Elevated; storage may be slow
Average VM boot time	170 seconds	Slow; may indicate resource or startup delays

3. Analysis of likely bottlenecks

A. CPU bottleneck

CPU utilization is 92%, indicating that the processor is heavily occupied.

Possible causes:

- Too many VMs running on the same host.
- CPU-intensive applications.
- Excessive virtual CPU allocation or CPU contention.

Corrective actions:

- Migrate some VMs to another host.
- Distribute workloads across multiple servers.
- Monitor CPU-ready time and processor usage.
- Add CPU capacity or upgrade the host if necessary.

B. Memory bottleneck

Memory utilization is 88%, meaning most of the available RAM is in use.

Possible causes:

- Too many VMs sharing the host.
- Applications consuming excessive memory.
- Memory overcommitment or swapping.

Corrective actions:

- Identify VMs consuming excessive memory.
- Allocate RAM according to actual workload requirements.
- Reduce unnecessary VM workloads.
- Add physical RAM or migrate VMs to another host.

C. Storage I/O bottleneck

Storage I/O latency is 35 ms. This indicates that storage requests may be taking too long, depending on the workload and storage system.

Possible causes:

- Slow storage devices.
- Heavy read/write activity.
- Multiple VMs competing for the same storage.
- Storage network congestion.

Corrective actions:

- Monitor disk latency, IOPS and throughput.
- Distribute storage-intensive VMs across storage resources.
- Use faster SSD or NVMe storage where appropriate.
- Optimize storage configuration and investigate network delays.

D. Slow VM boot time

The average VM boot time is 170 seconds, or 2 minutes and 50 seconds. This may indicate resource contention or delays during startup.

Possible causes:

- High CPU and memory utilization.
- Slow storage access during boot.

- Too many startup services or applications.
- Network or operating-system initialization delays.

Corrective actions:

- Reduce CPU, memory and storage contention.
- Disable unnecessary startup services.
- Investigate VM boot logs and storage performance.
- Optimize VM templates and distribute simultaneous startup workloads.

4. Recommended corrective action priority

Priority 1: Reduce resource contention

CPU usage is 92%. Check CPU-ready time and migrate heavily loaded VMs to less busy hosts.

Priority 2: Investigate storage latency

Check storage queues, IOPS and throughput to determine why latency has reached 35 ms.

Priority 3: Review memory usage

Identify memory-intensive VMs and check for swapping or memory pressure.

Priority 4: Optimize VM startup

Review boot logs and measure boot time again after addressing resource contention.

5. Conclusion

The VM host shows signs of possible resource bottlenecks, particularly high CPU utilization, high memory utilization and elevated storage latency. These conditions may contribute to the average VM boot time of 170 seconds. The cloud administrator should monitor resource usage, redistribute workloads, optimize storage and increase capacity where necessary. After applying the corrective actions, performance should be measured again to verify improvement.

Question 4: Software-Defined Datacenter (SDDC) Architecture (10 Marks)

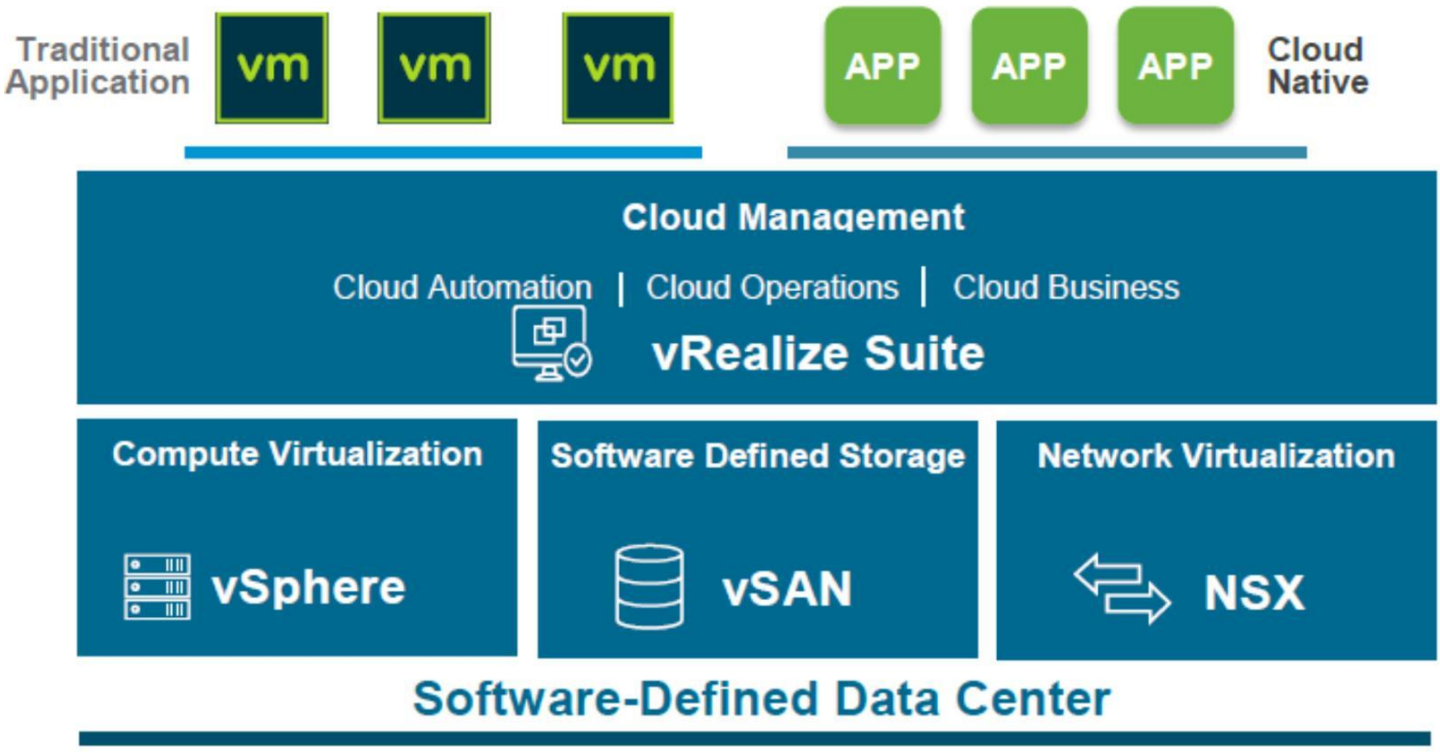
Question: Analyze how Software-Defined Datacenter architecture improves agility compared to a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

1. Introduction

A traditional datacenter uses physical servers, storage devices and network equipment that are often configured and managed separately. Making changes can require manual configuration and considerable time.

A Software-Defined Datacenter (SDDC) is a datacenter in which compute, storage and networking resources are virtualized and managed through software. It enables centralized management, automation and faster resource allocation.

2. Traditional datacenter vs SDDC



Feature	Traditional datacenter	Software-Defined Datacenter
Resource management	Mostly manual and hardware-oriented	Software-based and centralized
Compute	Physical servers are configured individually	Virtual machines and virtualized compute resources
Storage	Separate physical storage systems	Software-managed storage pools
Networking	Physical switches and routers configured manually	Software-defined networks and virtual switches
Scalability	May require manual hardware setup	Resources can be allocated through software
Agility	Slower changes and deployment	Faster provisioning and automation
Cost efficiency	Risk of underutilized hardware	Better resource pooling and utilization

3. How SDDC improves agility

Agility means the ability to respond quickly to changing business or application requirements.

A. Compute virtualization

Compute virtualization allows multiple virtual machines to run on physical servers through a hypervisor.

Benefits:

- Virtual machines can be created quickly.

- CPU and memory can be allocated based on workload requirements.
- Applications can be migrated between compatible hosts.
- Resources can be scaled without purchasing a separate server for every application.

  Example: If an online shopping application experiences increased traffic, administrators can allocate additional computing resources or deploy more application instances.

  B. Storage virtualization

  Storage virtualization combines storage resources from multiple devices into logical storage pools managed through software.

  Benefits:

- Storage capacity can be allocated when required.
- Storage management becomes simpler.
- Data can be replicated and protected using suitable storage policies.
- Storage resources can be utilized more efficiently.

  Example: If a database requires additional storage, an administrator can allocate capacity from a shared storage pool instead of manually installing and configuring a separate storage device.

  C. Network virtualization

  Network virtualization creates logical networks independently of the underlying physical network infrastructure. Technologies such as software-defined networking (SDN) enable programmable network control.

  Benefits:

- Virtual networks can be created quickly.
- Network policies can be applied centrally.
- Workloads can be isolated using network segmentation.
- Network configuration can be automated.

  Example: A cloud administrator can create a separate virtual network for a finance application without building an entirely new physical network.

4. Major advantages of SDDC

1. Rapid provisioning: Computing, storage and networking resources can be deployed quickly.
2. Automation: Repetitive administrative tasks can be automated.
3. Scalability: Resources can be expanded or reduced according to demand.
4. Improved utilization: Shared resources reduce the need for dedicated hardware for every workload.
5. Centralized management: Administrators can manage resources using common software tools.
6. Security: Policies and workload isolation can be applied consistently across virtual infrastructure.

5. Example of SDDC in a cloud environment

Consider a college that needs to deploy a student portal before admissions begin.

In a traditional datacenter, administrators might need to configure additional servers, storage and networking manually.

With an SDDC, administrators can provision virtual machines, allocate storage and configure virtual networks through software-based management tools. Resources can then be scaled according to student traffic.

This allows the college to deploy services faster and respond more easily to changing demand.

6. Conclusion

A Software-Defined Datacenter improves agility by virtualizing compute, storage and networking resources and enabling centralized, automated management. Compared with a traditional datacenter, it supports faster deployment, flexible scaling, improved resource utilization and consistent security policies. Therefore, SDDC is well suited to modern cloud environments where resources must be managed efficiently and quickly.

Question 5: Multi-Cloud Federation and Secure Identity Management (10 Marks)

Question: A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

1. Introduction

Multi-cloud computing is the use of cloud services from multiple providers, such as AWS, Microsoft Azure and Google Cloud.

Cloud federation allows different cloud providers to work together by establishing trust and enabling authorized users to access resources across cloud environments.

However, federation can fail when providers use different identity formats, authentication methods, user roles or security policies. These mismatches can prevent users from accessing services or may create security risks.

2. Causes of identity mismatches

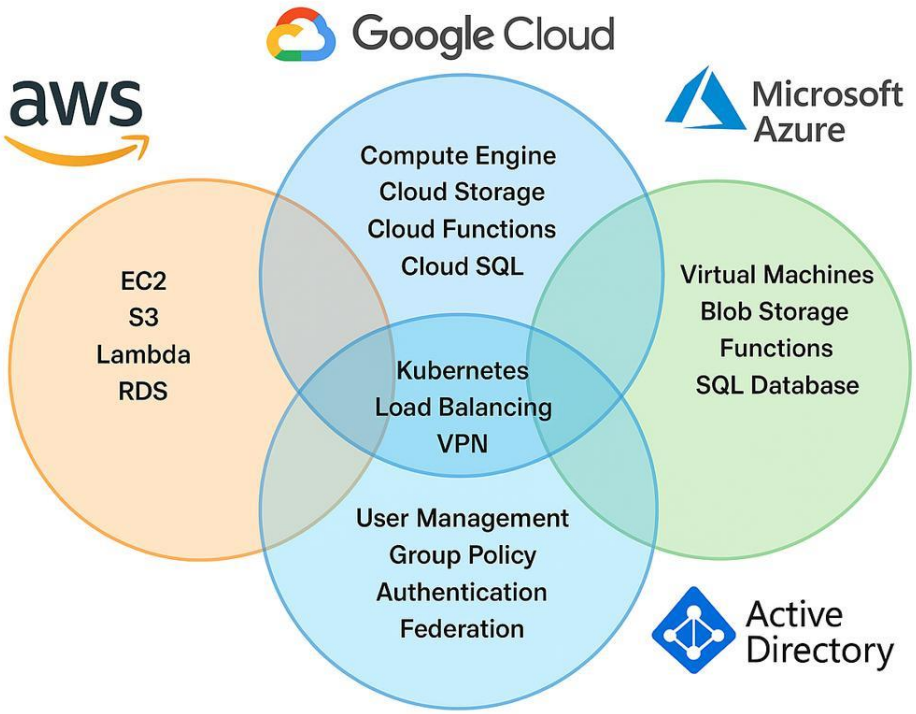
1. Different identity formats: Providers may use different usernames, user IDs or email addresses to identify the same user.
2. Different authentication methods: One provider may use single sign-on (SSO), while another requires separate authentication.
3. Role mismatches: A role in one cloud may not correspond to the correct permissions in another cloud.
4. Token incompatibility: Providers may expect different token formats, issuers, audiences or signing keys.
5. Inconsistent security policies: Password, multi-factor authentication and session policies may differ between providers.

Example

Suppose a university uses Azure for its student portal and AWS for its learning management system. A student successfully logs in to Azure but cannot access the AWS application because the AWS service does not recognize the user's identity or assigned role.

This is an identity federation problem.

3. Proposed secure identity design



The following architecture can help solve the problem.

A. Central Identity Provider (IdP)

Use a trusted central Identity Provider to authenticate users.

Examples include Microsoft Entra ID and other enterprise identity providers.

Functions:

- Maintains user identities.
- Authenticates users.
- Supports multi-factor authentication.
- Issues trusted authentication tokens or assertions.

B. Single Sign-On (SSO)

Implement SSO so users can authenticate once and access authorized applications across multiple clouds without repeatedly entering credentials.

For example, an authorized university employee can sign in once and access permitted services hosted on different cloud platforms.

C. Standard federation protocols

Use established protocols to enable secure communication between identity providers and cloud services.

- SAML 2.0: Exchanges authentication assertions between an identity provider and a service provider.
- OpenID Connect (OIDC): Provides an identity layer on top of OAuth 2.0 using standardized identity tokens.
- OAuth 2.0: Enables delegated authorization to access protected resources.

The selected protocol must be supported and correctly configured by the participating services.

D. Identity and role mapping

Create a common identity-mapping policy that maps user identities and roles between cloud providers.

For example:

User role	AWS permissions	Azure permissions
Administrator	Manage approved cloud resources	Manage approved Azure resources
Faculty	Access assigned applications and storage	Access assigned applications and storage
Student	Access learning resources	Access student portal resources

Permissions should be assigned according to each user's responsibilities. The same role name should not automatically grant identical permissions in every cloud.

4. Security and privacy measures

The following measures should be implemented:

1. Multi-factor authentication (MFA): Require additional verification for sensitive accounts and operations.
2. Least-privilege access: Give users only the permissions necessary for their work.
3. Secure token validation: Validate token signatures, issuer, audience, expiry and other required claims.
4. Encryption: Protect identity information and communications using appropriate encryption.

5. Token expiration and revocation: Use short-lived access tokens where appropriate and revoke access when necessary.
6. Audit logging: Record authentication attempts, access decisions and administrative changes.
7. Data minimization: Share only the identity attributes required by the receiving cloud service.
8. Account lifecycle management: Update or remove permissions when users change roles or leave the organization.

5. Implementation steps

1. Identify the identity systems and authentication protocols used by each cloud provider.
2. Select a trusted central identity provider.
3. Configure federation between the identity provider and each cloud.
4. Map user identities and roles to the correct cloud permissions.
5. Enable MFA, encryption and appropriate token-validation rules.
6. Test successful login, unauthorized access, expired tokens and revoked accounts.
7. Monitor logs and regularly review access policies.

6. Expected outcomes

The proposed design helps achieve:

- Consistent authentication across multiple clouds.
- Reduced need for separate user credentials.
- Fewer identity-mapping errors.
- Better control over user permissions.
- Improved privacy and auditability.
- Reduced risk of unauthorized access.

7. Conclusion

Multi-cloud federation can fail when identity formats, authentication protocols and authorization policies differ between cloud providers. A secure solution is to use a trusted identity provider, SSO, standard federation protocols such as SAML or OIDC, and carefully defined identity and role mappings. MFA, least-privilege access, token validation, encryption and audit logging further strengthen security and privacy.